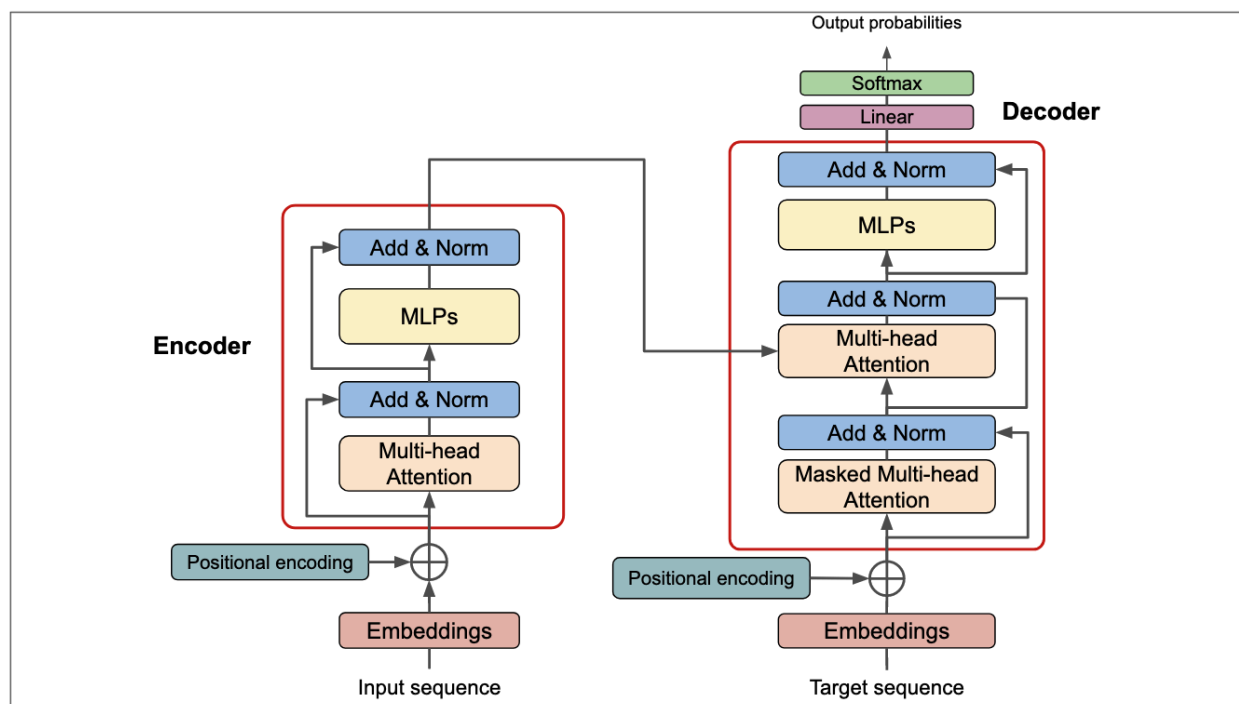
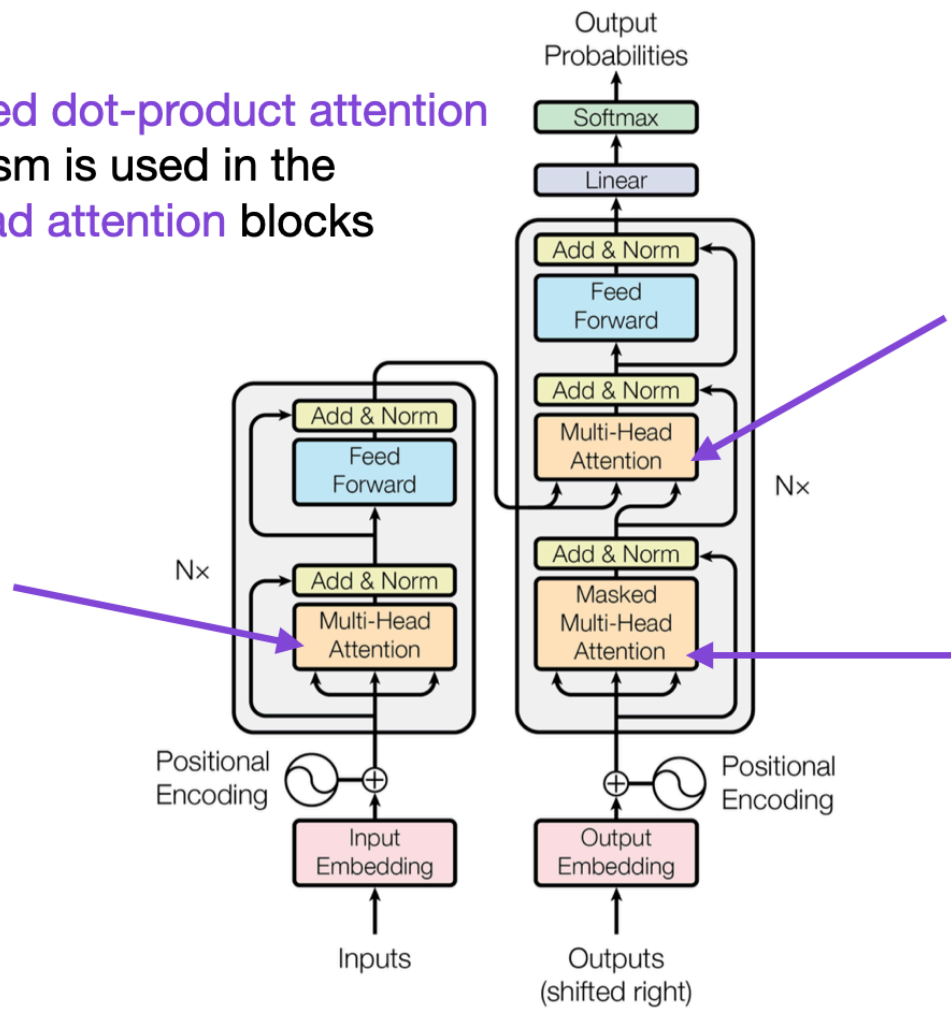




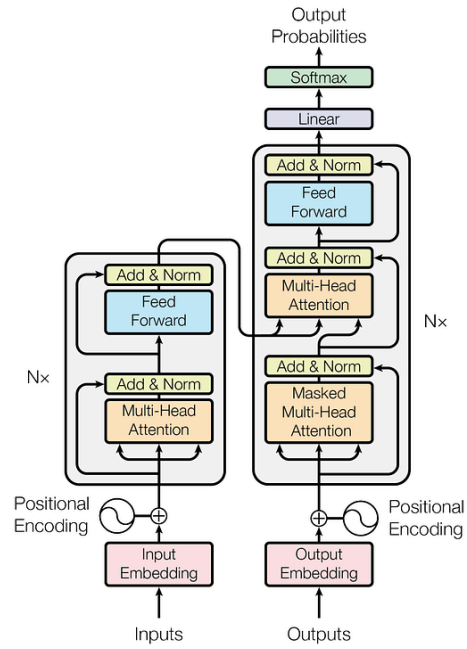
The **scaled dot-product attention** mechanism is used in the **multi-head attention blocks**



Source: "Attention Is All You Need" (<https://arxiv.org/abs/1706.03762>)

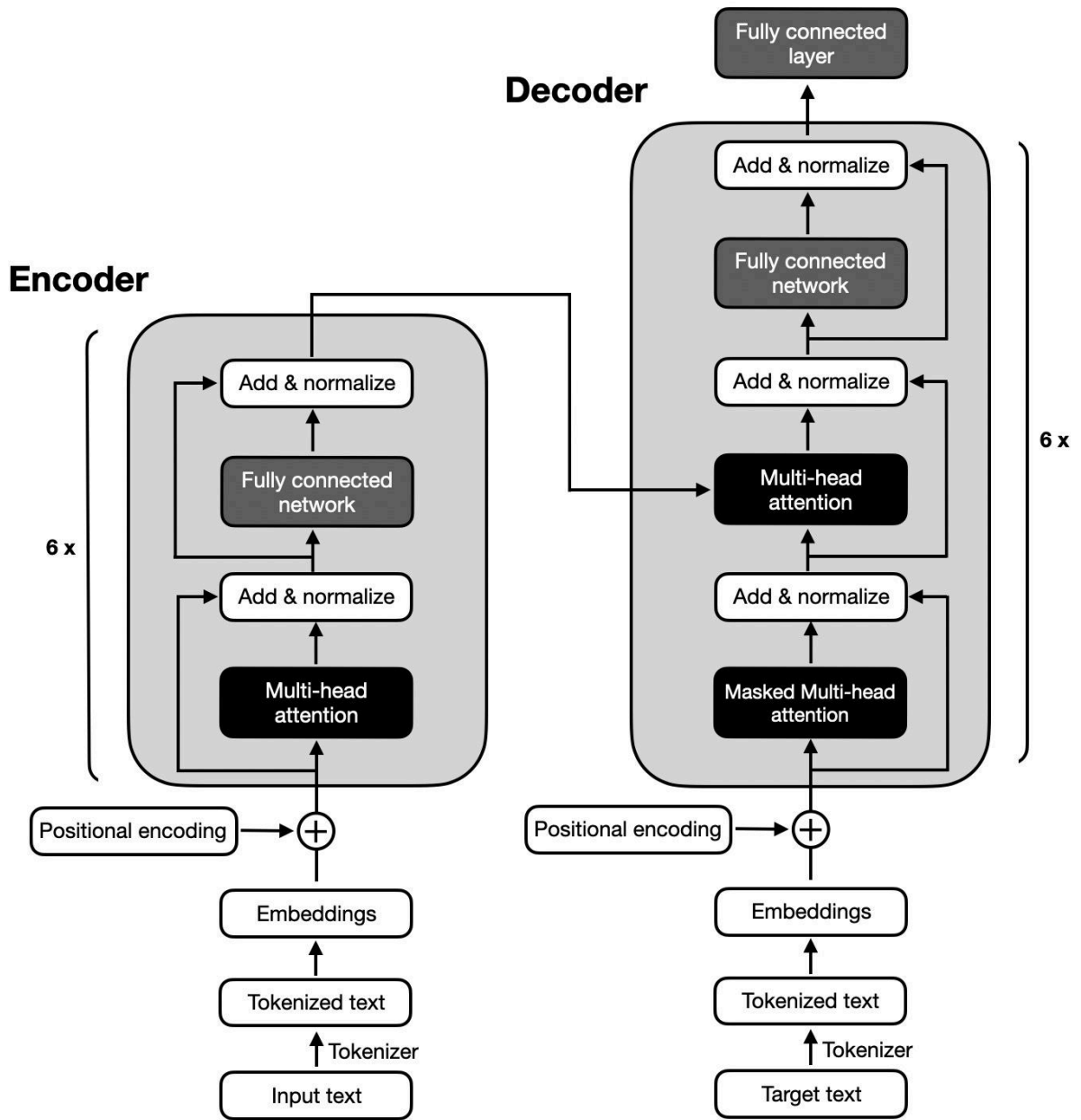
# BERT

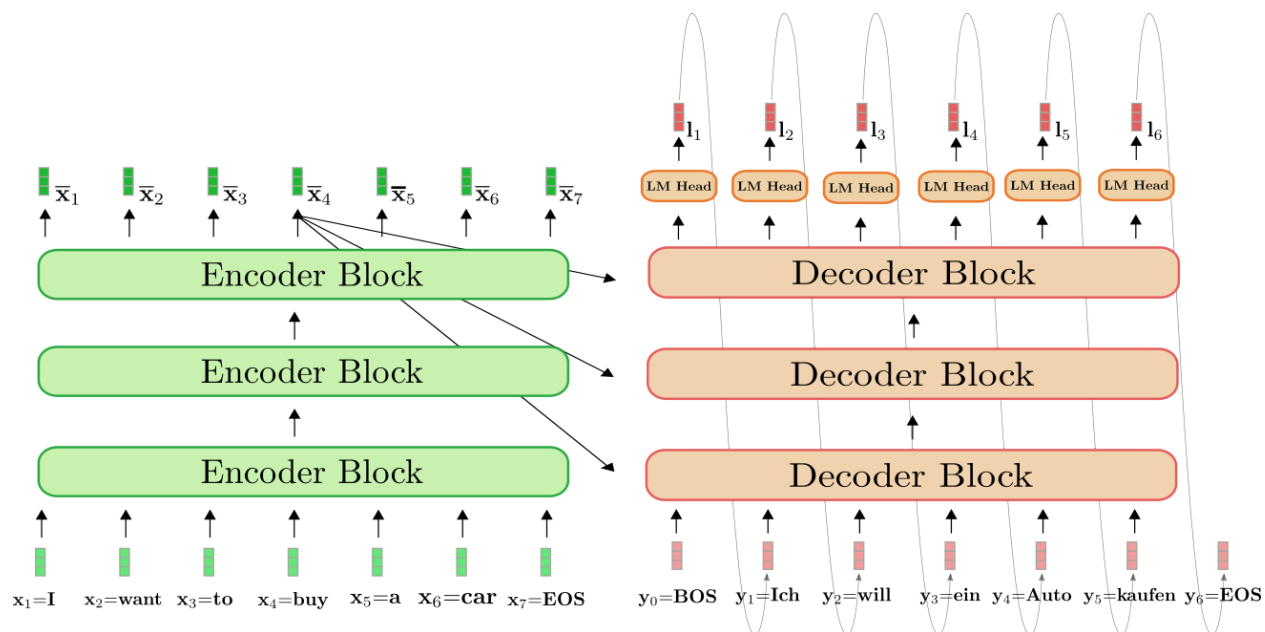
Encoder



# GPT

Decoder





## Transformer architecture

The Transformer architecture is a deep learning neural network design introduced by Vaswani et al. in the 2017 paper *Attention Is All You Need*. It replaces recurrence and convolution with self-attention mechanisms, enabling parallel sequence processing and long-range dependency modeling. This innovation became the foundation for large language models and multimodal AI systems across text, vision, and speech. ([Emergent Mind](#))

### Key facts

- **Introduced:** 2017 (Google Research)
- **Core mechanism:** Multi-head self-attention
- **Canonical structure:** Encoder-decoder stack (6+ layers each)
- **Computational form:** Scaled dot-product attention
- **Influential descendants:** BERT, GPT, T5, Vision Transformer (ViT)

### Architecture and operation

A standard transformer comprises an **encoder** that maps input tokens to contextual representations and a **decoder** that autoregressively generates outputs. Each layer includes multi-head self-attention, position-wise feed-forward networks, residual (skip) connections, and layer normalization. Self-attention computes pairwise interactions between all tokens, allowing

parallelized learning of relationships independent of distance. The architecture injects order information via sinusoidal or learned **positional encodings**.

## Mechanism of self-attention

For queries (Q), keys (K), and values (V) derived from input embeddings, attention weights are computed as

$$\left[ \text{softmax} \left( \frac{QK^{\text{top}}}{\sqrt{d_k}} \right) V \right]$$

where ( $d_k$ ) is key dimensionality. Multi-head attention runs this in parallel across several subspaces and concatenates outputs, enabling the model to capture syntactic, semantic, and long-range dependencies simultaneously. ([21medien](#))

## Advantages and scalability

Transformers eliminate the sequential bottlenecks of RNNs and LSTMs by processing entire sequences concurrently, which greatly accelerates training and enables scaling to trillions of parameters. The constant path length between tokens improves modeling of distant context, while architectural elements such as residual links and layer normalization enhance gradient stability.

## Impact and extensions

Since their introduction, transformers have redefined sequence modeling. Encoder-only forms (BERT, RoBERTa) dominate understanding tasks; decoder-only forms (GPT series, Claude, LLaMA) lead in text generation; and encoder–decoder hybrids (T5, BART) excel at translation and summarization. Variants now extend to **vision**, **audio**, **graph**, and **multimodal** data, inspiring innovations like sparse attention, rotary position embeddings, and mixture-of-experts scaling. ([MDPI](#))

